

# ECE 3113 - Energy Conversion

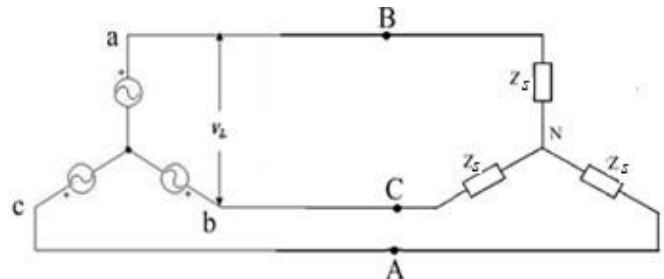
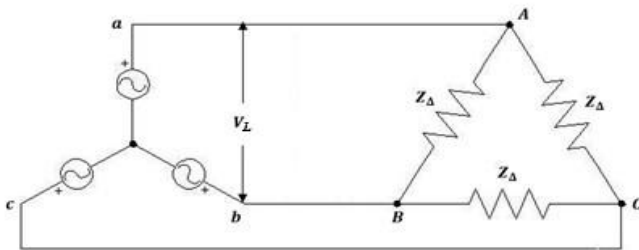
## Homework 3

(Due on Monday 10/16/2023)

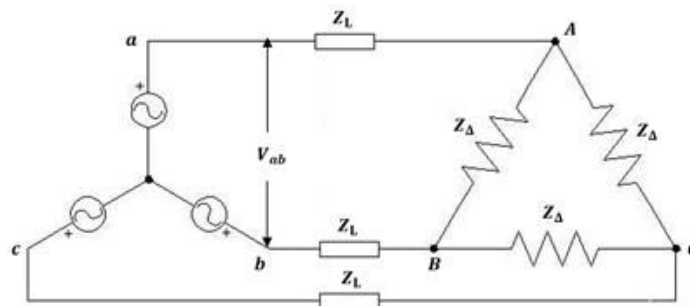
1.

- Phase voltages of a star connected alternator are  $V_{an} = 231 \angle 60^\circ \text{ V}$ ,  $V_{bn} = 231 \angle 180^\circ \text{ V}$ , and  $V_{cn} = 231 \angle -60^\circ \text{ V}$ . Determine the phase sequence of the system.
- In a balanced, positive sequence, star connected alternator  $V_{bn} = 231 \angle -70^\circ \text{ V}$ . Calculate  $V_{an}$  and  $V_{cn}$ .

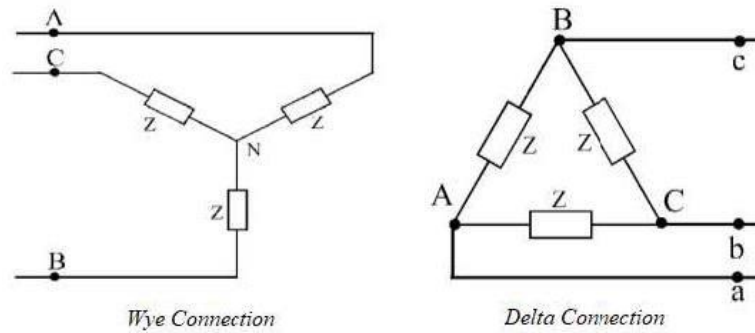
- In the circuits given below,  $Z_\Delta = 10 \Omega$ ,  $Z_s = 10 \Omega$ ,  $V_L = 230 \text{ V}$ . For each of the circuits below, find the rms values of:
  - Line current
  - Phase current and phase voltage of load
  - Phase voltage and phase current at the source



- A balanced, positive-sequence, wye-connected voltage source with  $V_{ab} = 300 \angle 0^\circ \text{ V}$  is connected to a balanced  $\Delta$  load with  $Z_\Delta = 30 \angle 30^\circ \text{ ohm}$  per phase. The impedance of the line between the source and the load is  $Z_L = 3 \angle 60^\circ \text{ ohm}$  per phase. Draw the per phase representation of the given 3 phase circuit.



4. Three similar coils, each having a resistance of  $20\ \Omega$  and an inductance of  $0.1\ \text{H}$  are connected first in star formation and then in delta formation to a 3 phase,  $60\ \text{Hz}$  supply with  $150\ \text{V}$  between lines. Calculate the total power absorbed and the line current in each case i.e. (i) wye, (ii) delta



5. In the balanced  $3\phi$  system shown, make a wye-connection of this balanced three phase circuit at blue box at the terminal.

